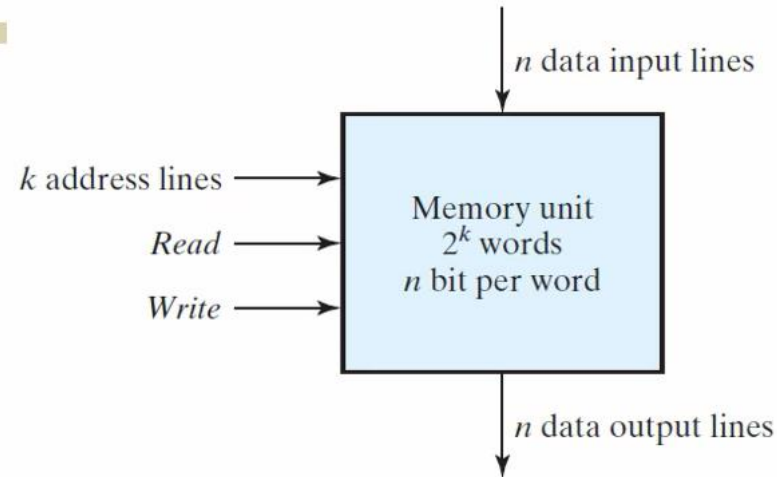


RAM: Write and Read Operations



To transfer a new word to be stored into memory:

1. Apply the binary address of the word to address lines.
2. Apply the data bits that must be stored in memory to the data input lines.
3. Activate the *write* input.

To transfer a stored word out of memory:

1. Apply the binary address of the word to address lines.
2. Activate the *read* input.

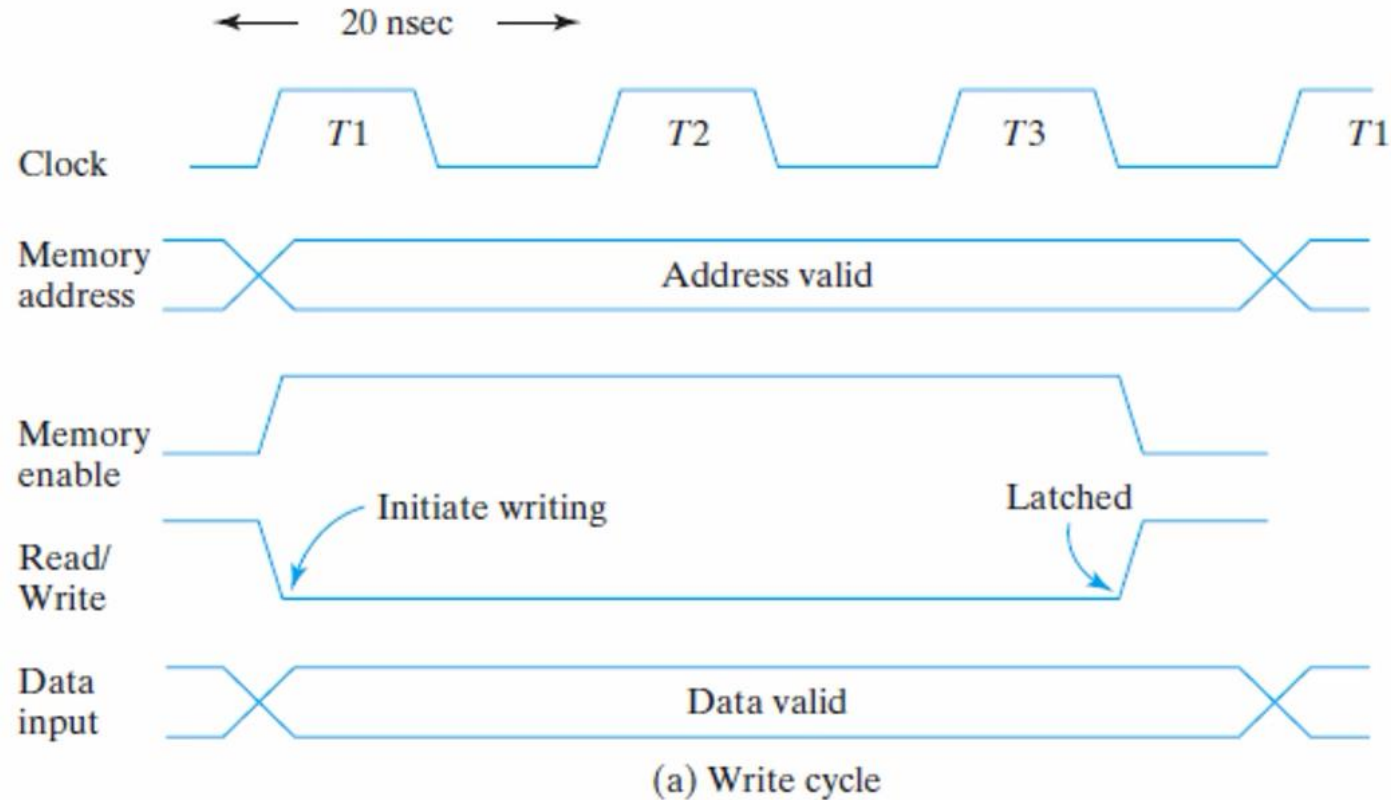
RAM: Write and Read Operations

Table 7.1

Control Inputs to Memory Chip

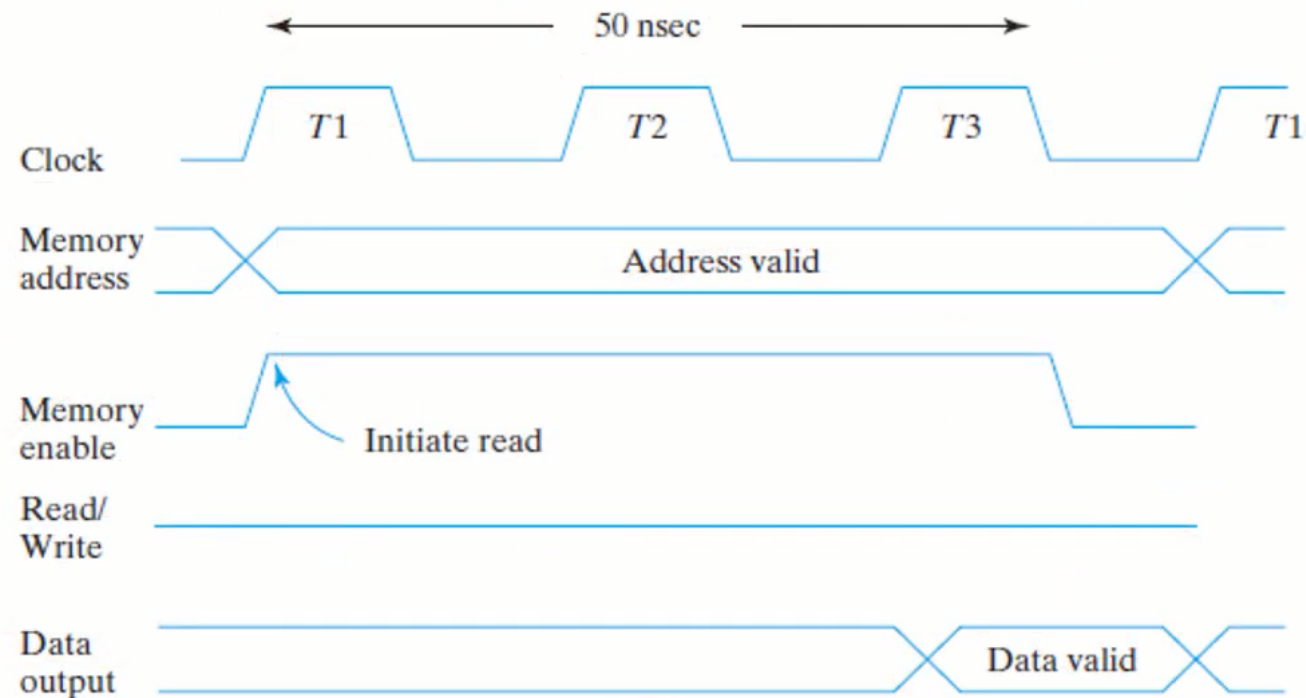
Memory Enable	Read/Write	Memory Operation
0	X	None
1	0	Write to selected word
1	1	Read from selected word

Write Operation in RAM



The memory enable switches to the high level and the Read/Write to the low level. The two control signals must be active for at least 50 ns.

Read Operation in RAM



(b) Read cycle

The memory enable switches to the high level and the Read/Write to the high level. The memory places the data of the word selected by the address into the output data lines in at most 50 ns from the time the memory enable is activated.